



Forests or trees? The effect of generating solutions to distant analogies on global–local processing

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Abstract

The present research was designed to examine the effect of solving distant analogies on global–local processing. In two experiments, participants generated solutions to near analogies (near condition), or distant analogies (distant condition), and then they were required to either complete the Kimchi–Palmer task (Experiment 1) or the Navon letter task (Experiment 2). The experimental results showed that participants who generated solutions to distant analogies scored higher on the Kimchi–Palmer task and had faster reaction times to global letters. These findings indicated that solving distant analogies could promote global processing.

Introduction

When people view the painting “The Starry Night” by well-known artist Van Gogh, sometimes they focus on the overall artistic mood of the painting, and sometimes they focus on the stars or the moon in the painting. That is, people can either attend to the Gestalt of a stimulus or pay more attention to its details (Navon, 1977; Schooler, 2002). Studies have shown that attention to global or local characteristics of a stimulus could activate global or local processing (e.g., Burgoon et al., 2013). When global processing is activated, individuals’ attention tends to focus on the global characteristics of the stimulus, which widens the activation range of stimulation-associated concept nodes in the semantic network to facilitate information integration into a superordinate concept or existing knowledge structure. On the contrary, when local processing is activated, individuals are likely to concentrate on the local characteristics of the stimulus, which narrows the activation range of stimulation-associated concept nodes in the semantic network to facilitate the activation of specific subordinate concepts in the semantic network (Burgoon et al., 2013; Collins & Loftus, 1975; Henderson, 2013).

Across disciplines, studies have demonstrated that global processing or local processing has different effects on cognitive processing in individuals. Research has established a close relationship between global processing and the regulation of negative emotions (Gu et al., 2017; Ji et al., 2019), reading comprehension (Franceschini et al., 2021), creative problem-solving (Zmigrod et al., 2015), interpersonal forgiveness (Mok & De Creme, 2015), the promotion of prosocial behavior (Mukherjee et al., 2018), reducing change-localization abilities (Wilford & Wells, 2010), reducing the conscious accessibility of their parts (Poljac et al., 2012), and improving change detection but impairing change identification (Mathis & Kahan, 2014). Local processing, however, is closely associated with the facilitation of single-word reading as well as text reading rate (Viterbo et al., 2020), impairing the fast processing of the letter string useful for lexical reading (Franceschini et al., 2021), the perception of sadness emotion (Srinivasan & Hanif, 2010) and categorizing information based on taxonomic relations (Guest et al., 2016).

Which factors elicit global or local processing? Research has demonstrated that emotions play an important role in determining whether individuals are inclined towards global or local processing (e.g., Baumann & Kuhl, 2005; Gasper & Clore, 2002; Gable et al., 2015). On one hand, Gasper and Clore (2002), examining the potency dimension of emotions, found that positive emotions could make individuals tend to process information globally. In the study, the researchers elicited positive and negative emotions within both groups of participants, followed by the administration

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of the Kimchi–Palmer task. In this task, participants were shown a hierarchical “standard” figure in which the global shape was made up of several local shapes (e.g., a square made of triangles). The participants were then asked to choose which of two “comparison” hierarchical figures (squares or triangles) best matched a standard figure. One of the comparison figures matched the standard at the global level and one at the local level. The total number of test triads in which the global comparison shape was selected was calculated for each participant, resulting in a global score. Higher scores indicated that participants favored global processing. The results found that compared with the negative emotion group, the positive emotion group scored higher on the Kimchi–Palmer task. Baumann and Kuhl (2005) further found that positive emotions could enhance the adaptability of transitioning between global–local processing. On the other hand, Gable et al. (2015), exploring the motivational dimension of emotions, also discovered that emotions with high motivational intensity (e.g., anger) predispose individuals to process information locally. Nonetheless, some studies do not appear to establish a direct correlation between emotion and global–local processing (De Luca et al., 2023). Furthermore, research has also revealed that an individual’s cultural background significantly influences their tendency to process information either globally or locally (e.g., Li et al., 2023b; Masuda & Nisbett, 2001; Wan et al., 2016). Masuda and Nisbett (2001) discovered that East Asians tend to prefer global processing, whereas Euro-Americans have a preference for local processing. From a cultural exchange perspective, Wan et al. (2016) observed that multicultural experiences tend to encourage members of the majority group to process information globally. Conversely, Li et al., (2023b) found that multicultural experiences can predispose minority group members to process information locally. Recent studies have indicated that an individual’s thinking style has an impact on their tendency to process information either globally or locally (e.g., Liberman & Förster, 2009a; Dijkstra et al., 2012; Li et al., 2017). Liberman and Förster (2009b) found that compared to a low-level construal mindset, a high-level construal mindset could make individuals tend to process information globally. Liberman and colleagues initiated a high-level or low-level construal mindset for two groups of participants and then completed the Navon letter task. This task indexes how fast people can process global vs. local characteristics of hierarchically constructed visual stimuli (e.g., larger letters made of smaller letters). The elements at the two different levels can either be the same or different. Participants are usually directed to attend to either the global or the local level and to identify the stimulus at that level as quickly as possible. The results found that participants who activated a high-level construal mindset responded more quickly to global letters. Dijkstra et al. (2012) discovered that intuition tends to facilitate

global processing, whereas deliberation tends to encourage local processing. Li et al. (2017) also found the tendency of unconscious thought toward global processing.

In addition to the aforementioned factors, studies have indicated that people likely adopt global or local processing that allows their current need fulfillment (e.g., Hansen & Steinmetz, 2019; Kazakova et al., 2015; Marguc et al., 2011; Wronska et al., 2018). Marguc et al. (2011) discovered that participants, upon encountering a physical obstacle while navigating a maze, triggered an “if obstacle, then initiate global processing” response to overcome the hindrance. In a separate study, Hansen and Steinmetz (2019, Experiment 3) found that when products align with participants’ desired goals, the tendency is for individuals to assess them relying on local (i.e., single reviews) rather than global product information (i.e., overall ratings). Kazakova et al. (2015) similarly discovered that, in contrast to sequential media consumption, engaging in multiple distinct media tasks concurrently fosters a propensity for more localized processing in subsequent tasks. Adding to these findings, Wronska et al. (2018) have more recently demonstrated that, after completing a creative thinking task, participants exhibit a preference for processing information globally as opposed to after completing an analytical thinking task. The researchers believe that it is a result of adjusting the way cognitive processing to the task requirements (Marguc et al., 2011; Wronska et al., 2018). Additionally, solving problems by identifying relationships and drawing connections between disparate phenomena or objects is one of the cognitive tasks that people regularly encounter. For instance, in everyday life, it involves explaining to a child why a teacher might liken the stem of a plant to a straw, or understanding and elucidating metaphors such as “the Internet is like a highway” to others. In the realm of science and innovation, George de Mestral examined the relationship between burrs and his dog’s fur, ultimately leading to the development of Velcro based on this connection. Similarly, Claude Shannon’s Masters thesis extended the structure of Boolean Logic to electric circuits, which was central to the development of the computer chip. Research has shown that one basic mechanism for perceiving and using relational similarity is comprehending and solving analogies, which embodies humankind’s remarkable mental aptitude (Gentner & Maravilla, 2017; Vendetti et al., 2014). Nevertheless, the impact of solving analogies on the way that individuals process information has yet to be explored in research.

An analogy is a mapping from a source domain to a target domain, it is a central cognitive process that underlies the human capacity for learning and creativity (Gentner & Maravilla, 2017; Holyoak & Thagard, 1997). However, not all analogies are created equal. The amount of overlap between domains has been used to characterize an analogy along a continuum from “near” to “distant” (Gentner, 1983;

Ward, 1998). That is, the source and target domains can be perceptually and/or conceptually similar in a near analogy, or quite different in a distant analogy. For instance, if you are walking near the harbor and you observe that catching salmon with a drift net is similar to catching shrimp with a trawl net, you have made a near analogy. If, in walking from the harbor back to your dorm, you pass the Math building and observe that using nets to catch fish is similar to using prediction intervals to predict future sample characteristics, you have made a far analogy.

Research has shown that solving distant analogies affects an individual's cognitive performance on subsequent tasks. The research found that compared with solving near analogies, solving distant analogies is beneficial for discovering structural relational similarities between things (Vendetti et al., 2014). Vendetti et al. asked one group of participants to solve a series of near analogies and the other group to solve a series of distant analogies. Whereas near analogies can be solved by matching identical relations (e.g., the furnace burns coal as a woodstove burns wood), distant analogies require evaluating or generating a more abstract relation that bridges the domains (e.g., the furnace burns coal as the stomach “burns” food). After the two groups of participants had solved all the analogies, they were asked to complete the picture-mapping task. The experimental results found that participants solving distant analogies gave more structural relational responses. Studies have also shown that compared to solving near analogies, solving distant analogies facilitates innovation (Dunbar & Blanchette, 2001), understanding (Barnett & Ceci, 2002), a reduction in belief-based responses in transitive inference (Andrews & Vann, 2019), relational integration (Andrews & Bohadana, 2018), creative processing (Green, 2016; Srinivasan et al., 2018), analogical retrieval (Goldwater & Jamrozik, 2019) and categorize information based on thematic relations (Li et al., 2023a), among others. A question arises then, as to whether solving distant analogies could promote global processing or local processing.

Research has indicated that solving distant analogies could activate a high level of construal (e.g., Chaxel, 2015; Green et al., 2012). Chaxel (2015) found that solving distant analogies allows individuals to create high-level explanatory statements about people's behavior. Green et al. (2012) found that solving distant analogies increased the activity of the left frontal brain region. The activity of this brain region is closely related to abstract construal (Gonen-Yaacovi et al., 2013). Additionally, Research has also indicated a high level of construal can activate holistic thinking (e.g., Liberman & Förster, 2009b; Smith & Trope, 2006). Smith and Trope (2006) found that increasing psychological distance by initiating a high power causes individuals to tend to perceive structure to extract the gist. Liberman and Förster (2009b) found that thinking about a more distant future, spatial

location, or social relationships makes individuals recognize Navon global letters faster. However, when holistic thinking is activated, people tend to prefer to attend to the linkages among objects and orient towards the context or field as a global, instead of focusing on the unique properties of discrete objects (Ng et al., 2021). Based on these findings, we hypothesize that solving distant analogies could promote global processing.

In the current study, solving either distant analogies or near analogies was manipulated through the type of verbal analogy task they completed. In Experiment 1, participants solved four-term verbal analogy problems, with half of the participants receiving near analogies and the other half receiving distant ones. They then completed a task to measure global–local processing. If solving distant analogies could promote global processing, the experimental results showed that participants who generated solutions to distant analogies scored higher on the Kimchi–Palmer task (Experiment 1) and had faster reaction times to global letters (Experiment 2). Furthermore, we have ruled out the possibility that these results could be attributed to the notion that solving near analogies inhibits individuals process information globally.

Experiment 1

In Experiment 1, global–local processing was measured using the Kimchi–Palmer task (e.g., Gasper & Clore, 2002; Gable & Harmon-Jones, 2008). If solving distant analogies could promote global processing, then the score of the distant analogy group should be significantly higher than that of the near analogy group in the Kimchi–Palmer task.

Methods

Participants and design

We determined the sample size based on effects sizes found in research which found an effect of $d=0.88$ by Huntsinger et al. (2010) and another research which found an effect of $d=0.31$ by Gable and Harmon-Jones (2008). We enter the average of these two effect sizes ($d=0.59$), and a desired power of $1-\beta=0.80$, an alpha level of $\alpha=0.05$ into the program G*Power (Faul et al., 2007) resulted in a total sample size of 94. Because we sought to higher power, a total of 130 Chinese university students (63 male and 68 female, mean age = 22.09 years) in Experiment 1. Among these participants, 65 (32 male and 33 female) were assigned to the distant analogy group and 65 (31 male and 34 female) to the near analogy group. Everyone signed the informed consent form and received corresponding course credits.

The experiment employed a between-participants single-factor design. The question type of the analogy task was a between-participants variable, and the dependent measurement variable was the participant's score on the Kimchi–Palmer task.

Materials and procedure

First, we asked all participants to complete the verbal analogy task. The materials for the verbal analogy task were identical to the materials used by Green et al. (2010) and Vendetti et al. (2014). Participants were randomly assigned to one of two groups, with one group completing 40 distant analogy problems, and the other group completing 40 near analogy problems. The sequence of presentation of analogy problems was randomly generated by the computer. Each analogy problem was presented in the format of A:B::C:?, with letters representing words and “?” representing a missing D term. Each trial began with a 500-ms fixation cross, followed by the presentation of the A:B and C: ? pairs (The A:B pair appeared on the computer monitor above the C: ? pair.). Participants were told to write their answers on the answer sheet in front of them as soon as they thought of them, then press the space bar to move on to the next problem. Responses (D terms) were considered correct if they matched the solutions provided by Green et al. (2010) or if the D terms formed a good analogy when combined with the A, B, and C, terms.

After completing the analogy task, both groups were asked to complete the Kimchi–Palmer task (KPT, Kimchi & Palmer, 1982). Referring to previous studies, in the task, 24 items were presented on the computer screen in a sequence that was randomly generated by the computer (e.g., Kazakova et al., 2015; Dijkstra et al., 2017). The hierarchical shapes in each item consisted of small circular, square, or triangle shapes (local level) that formed a larger square or triangle (global level). On each item, a standard figure is presented at the top of the page with two comparison figures presented below (see Fig. 1). One of the comparison figures has the same global shape as the standard figure and different local details; the other was the other way around. Participants needed to determine which comparison figure was more similar to the standard figure. Participants were asked to press the “F” or “J” key to indicate their “first and most immediate impression” as to which of the two comparison figures in each triad best matched the standard figure. The total number of test triads in which the global comparison shape was selected was calculated for each participant, resulting in a global score. Higher scores indicated a preference for global processing among participants. Half of the comparative figures in the task that have a global shape similar to the standard figure were on the left, and half were on the right. At the end of the experiment, participants were

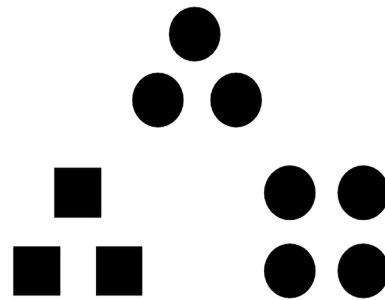


Fig. 1 Examples of the Kimichi and Palmer tasks

asked about the purpose of the experimental task, and only seven participants (5.38%) guessed the real purpose of completing the Kimchi–Palmer task.

Results and discussion

We calculated the accuracy of solving analogy problems for participants in the near analogy and distant analogy groups. Analysis using independent samples t-test revealed that the correct rate of participants in the near analogy group ($M=0.77$, $SD=0.09$) was significantly higher than that of participants in the distant analogy group ($M=0.67$, $SD=0.13$), $t(109)=5.39$, $p<0.01$, Cohen's $d=0.41$. These results replicate the findings of Green et al. (2012) and Vendetti et al. (2014), which suggest that the manipulation of analogy problem types in this study is effective.

We calculated the average score of the near analogy and distant analogy groups on the Kimchi–Palmer task (see Fig. 2). With the rate of correctly solving analogies as the covariate, analysis using One-way ANOVA revealed that the average score of participants in the distant analogy group ($M=19.12$, $SD=2.55$) was significantly higher than that of participants in the near analogy group ($M=11.81$, $SD=2.44$), $F(1, 128)=282.51$, $p<0.001$, $\eta_p^2=0.69$. The experiment results indicated that solving distant analogies could promote global processing.

Experiment 2

Research has shown that although the Kimchi–Palmer task and the Navon letter task are both classic tasks that measure global and local processing, they each measure a unique aspect (Dale & Arnell, 2013). In Experiment 2, we transformed the task of measuring global–local processing into the Navon letter task (e.g., Gable et al., 2015; Li et al., 2023b). In addition, previous research has shown that East Asians exhibit a stronger global precedence effect (e.g., Nisbett & Miyamoto, 2005). Thus, the results of Experiment 1 may be since solving near analogies inhibits individuals

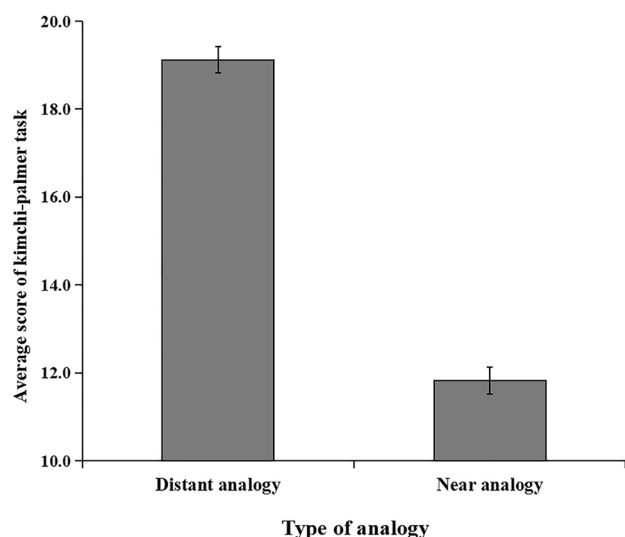


Fig. 2 The average score of the near analogy and distant analogy groups on the Kimchi–Palmer task (Error bars represent the average standard errors)

process information globally, rather than the fact that solving distant analogies could promote global processing. To rule out this possibility, a control group was also included in Experiment 2. If solving distant analogies could promote global processing, compared to the near analogy and control groups, participants in the distant analogy will react more quickly to global letters.

Methods

Participants and design

As in Experiment 1, we determined the required sample size based on the average effect size found in the related literature (e.g., Dale & Arnell, 2015; Liberman & Förster, 2009b). We enter the average of these two effect sizes ($f=0.46$), and a desired power of $1 - \beta=0.80$, an alpha level of $\alpha=0.05$ into the program G*Power (Faul et al., 2007) by specifying the number of groups with 3 and the number of measurements with 2, resulted in a total sample size of 50. Because we sought to higher power, a total of 102 Chinese university students (53 male and 49 female, mean age = 20.86 years) in Experiment 2. Among these participants, 34 (17 male and 17 female) were assigned to the distant analogy group, 34 (18 male and 16 female) to the near analogy group, and 34 (18 male and 16 female) to the control group. Everyone signed the informed consent form and received corresponding course credits.

A 3 (problem type of analogy task: near, distant, and control) $\times$ 2 (task type: global letter and local letter) two-factor mixed design was utilized in this experiment. The problem

type of analogy task was a between-participants variable, and the task type was a within-participants variable. The dependent measurement variable was the participants' reaction time in the Navon letter task.

Materials and procedure

Experiment 2 used the same analogy task as Experiment 1. As in Experiment 1, we asked one group of participants to solve 40 distant analogy problems and another group of participants to solve 40 near analogy problems. In addition, a control group was included in Experiment 2, and participants assigned to this group rated the pleasantness of 62 trait adjectives from Anderson (1968). Half of the adjectives were positive (e.g., Generous), and the other half were negative (e.g., Boastful). The presentation order of the adjectives was randomly generated by the computer. Participants were asked to rate each adjective's pleasantness on a scale from 1 to 7, where 1 represented very unpleasant, and 7 represented very pleasant.

After completing the analogy task (for the distant and near analogy groups) or word scoring task (for the control group), participants were asked to complete the Navon letter task adapted from Navon (1977). Referring to previous research, in this task, participants were presented with hierarchical stimuli in which the ignored dimension (global or local) was always neutral with respect to the response (Gable et al., 2015; Li et al., 2023a, b). Each stimulus included an image of a large letter (global letter) that was composed of many small letters (local letter) (see Fig. 2). The global letters (70 $\times$ 50 mm) were twenty times as large as the local letters (3.5 $\times$ 2.5 mm). An empty screen for 500 ms, followed by a hierarchical letter was displayed on the screen. If the participants don't respond within 2000 ms after a hierarchical letter appears, the screen will automatically display the next hierarchical letter. In the global letter identification task, participants were asked to determine whether the global letter that was composed of local letters was an H or an L. In the local letter identification task, participants were asked to determine whether the local letters that formed the global letter were Hs or Ls. If the global or local letter is "H", press "F"; if it is "L", press "J". For both the global and local letter identification tasks, all participants underwent a 10-trial practice session before the official task began. Only those who achieved an accuracy rate exceeding 90% during the practice were permitted to proceed to the formal task. In the formal task, 160 global trials and 160 local trials were presented. Half of the participants in each group were asked to complete the tasks involving identification of the global letters first, followed by the local letter identification tasks; whilst the remaining half were asked to work on the local letter identification tasks before the global letter identification tasks. Similarly, at the end of the experiment, participants

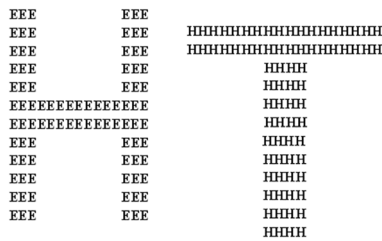


Fig. 3 Examples of the Navon letter task

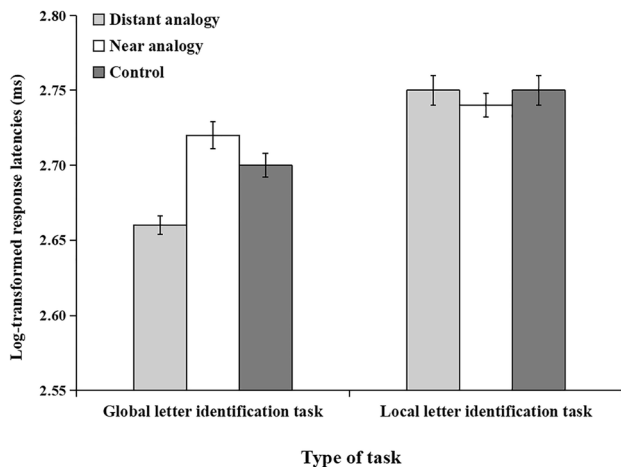


Fig. 4 The mean response latencies in milliseconds of the near analogy, distant analogy, and control groups in the Navon letter task (error bars represent the average standard errors)

were asked about the purpose of the experimental task, and only four participants (3.96%) guessed the real purpose of completing the Navon letter task (see Fig. 3).

Results and discussion

We also calculated the accuracy of solving analogy problems for participants in the near analogy and distant analogy groups. Analysis using independent samples t-test revealed that the correct rate of participants in the near analogy group ($M=0.73$, $SD=0.10$) was significantly higher than that of participants in the distant analogy group ($M=0.59$, $SD=0.13$), $t(66)=5.00$, $p<0.001$, Cohen's $d=0.52$. These results replicate the findings of Green et al. (2012) and Vendetti et al. (2014), which suggest that the manipulation of analogy problem types in this study is effective.

Before data analysis, logarithmic transformation was performed on the correct response times for each participant, and the average values of the Navon letter task were calculated for each participant (Beilock et al., 2007; Kühnen & Oyserman, 2002; Wan et al., 2016)(see Fig. 4). We used a paired-samples t-test to analyze and found that participants' response time to the Navon global letter identification task

($M=2.69$, $SD=0.05$) was noticeably faster than the response time to the Local letter identification task ($M=2.75$, $SD=0.05$), $t(66)=10.27$, $p<0.001$, Cohen's $d=1.01$. This result was consistent with the findings of Navon (1977).

Above all, with the rate of correctly solving analogies as the covariate, using 3 (problem type of analogy task: near, distant, and control) $\times$ 2 (task type: global letter and local letter) repeated measurement variance analysis (RMANOVA) revealed that the interaction between the problem type of analogy task and task type was significant, $F(2, 99)=15.78$, $p<0.001$, $\eta_p^2=0.25$. Post-hoc tests with Bonferroni corrections for multiple comparisons showed that for the global letter identification task, the reaction time of the participants in the distant analogy group ($M=2.66$, $SD=0.04$) was notably faster than those of participants in the near analogy ($M=2.72$, $SD=0.05$), $p<0.001$, and control groups ($M=2.70$, $SD=0.04$), $p<0.001$. Furthermore, the differences between the near analogy and control groups were not significant, $p=0.96$. As for the local letter identification task, the differences in reaction time between the near analogy ($M=2.74$, $SD=0.05$) and distant analogy groups ($M=2.75$, $SD=0.06$) were not significant, $p=0.95$; those in reaction time between the distant analogy and control groups ($M=2.75$, $SD=0.05$) were also not significant, $p=0.96$. Furthermore, the differences between the near analogy and control groups were not significant, $p=0.53$. This result suggested again that solving distant analogies could promote global processing.

General discussion

The present study manipulated the type of analogy task to investigate whether solving distant analogies could promote global processing. The results showed that compared to participants who solved near analogies, participants who solved distant analogies scored higher on the Kimchi–Palmer task (Experiment 1); and compared to control participants, participants who solved distant analogies had faster responses in the global letter identification task (Experiment 2). The result suggests that solving distant analogies could promote global processing.

This study expanded the existing research in which people likely adopt global and local processing that allows the fulfillment of their current needs. Research suggests that how individuals process information is flexible, contingent upon the situation, rather than being entirely constant (e.g., Love et al., 1999). Previous studies have discovered that overcoming encountered barriers, achieving motivationally relevant goals, performing several distinct media tasks simultaneously, and solving creative problems all affect how individuals process information globally or locally (Hansen & Steinmetz, 2019; Kazakova et al., 2015; Marguc et al., 2011;

Wronska et al., 2018). The present study further found that solving distant analogies could promote global processing.

This study also extended research on the effect of solving distant analogies on individual cognitive processing. Previous studies have shown that solving distant analogies facilitates the reduction in belief-based responses in transitive inference, relational integration, creative processing, analogical retrieval, and the discovery of structural relation similarity (e.g., Andrews & Vann, 2019; Andrews & Bohadana, 2018; Chaxel, 2015; Green, 2016; Vendetti et al., 2014). The present study further found that solving distant analogies could promote global processing. Research in the field of global–local processing has shown that global processing facilitates creative problem-solving (Zmigrod et al., 2015) and the discovery of structural relationships (Li et al., 2019). Thus, future research could further examine whether solving distant analogies facilitating creative processing and relational processing is due to the activation of global processing.

We believe that the reason why solving distant analogies could promote global processing may be that solving distant analogies could activate a relational mindset. Research has shown that prioritizing attention to the structural relations among elements is an important factor in explaining the relative precedence of global and local processing (e.g., Love et al., 1999). In particular, giving precedence to the structural relationships between elements promotes global processing. Several studies have underscored that solving distant analogies necessitates continuous exploration and evaluation of structural relations that bridge disparate domains (Chaxel, 2015; Li et al., 2023a, b; Vendetti et al., 2014). Consequently, individuals are likely to cultivate a relational mindset as they engage in solving distant analogies (e.g., Holyoak & Thagard, 1997; Simms & Richland, 2019). When a relational mindset is activated, people tend to seek out and prioritize relational information. Therefore, solving distant analogies could promote global processing. An additional data analysis we conducted seems to support this interpretation to some extent. In Experiment 1, we conducted a correlation analysis between the participants' correct rate of solving distant analogies in the distant analogies group and their scores on the Kimchi–Palmer task. The analysis revealed a significant positive correlation between the correct rate of solving distant analogies and Kimchi–Palmer task scores, $r(63) = 0.46$, $p < 0.001$; More importantly, regression analysis further found that the correct rate of solving distant analogies could significantly and positively predict the score of the Kimchi–Palmer task, $\beta = 0.46$, $t(63) = 4.08$, $p < 0.001$. Likewise, in Experiment 2, a noteworthy negative correlation emerged between the participants' correct rate of solving distant analogies and their response time to the global letter, $r(32) = -0.56$, $p < 0.01$, while the correlation between reaction time to the local letter

was less pronounced, $r(32) = -0.21$, $p = 0.22$. Using the methodology outlined by Eid et al. (2011), a comparison of correlations from dependent samples showed that the correlation between the correct rate of solving distant analogies and the global letter response time was significantly stronger than the correlation with the local letter response time, $z = -2.07$, $p = 0.02$. The regression analyses further substantiated this relationship, indicating that the rate of correctly solving distant analogies significantly and negatively predicted participants' response rate to the global letter, $\beta = -0.46$, $t(32) = -3.83$, $p < 0.01$. The results of the analyses from either Experiment 1 or Experiment 2 indicated a clear trend in which the better performance in solving distant analogies, individuals demonstrate a stronger propensity for processing information globally. This likely implies that the phenomenon in which participants in the distant analogy group tend to process information globally stems from the transfer of cognitive strategies. Of course, despite our attempts to explore the mechanisms involved through additional analyses, further examination by future studies is needed. In addition, limited by the experimental paradigm, we cannot arbitrarily assume that this shift did not occur automatically and rapidly, all of which still need to be further examined in future studies.

We believe that the absence of a significant difference in response rates between participants in the near analogy group and those in the control group, who completed the Navon letter task, may be attributed to the version of the Navon letter task employed in this study. Previous research has shown the impact of the saliency of global and local letters on the assessment of local processing (Dale & Arnell, 2013; Kimchi & Palmer, 1982). Specifically, the density of local elements constituting the global shape can influence the speed at which individuals recognize these local elements. Compared to the version in which the global and local levels were roughly equal in salience and the global letters (70×50 mm) were ten times as large as the local letters (7×5 mm) used by Dale and Arnell (), in the Navon letter task used in our study, the local letters comprising the global letter were marginally denser [The global letters (70×50 mm) were twenty times as large as the local letters (3.5×2.5 mm)], which might have somewhat influenced the speed at which participants recognized these local letters. Unlike solving distant analogies, solving near analogies needs to focus on discovering perceptual or conceptual surface similarities between things, which seems to be related to local processing (e.g., Li et al., 2023a). Future research should further examine whether solving near analogies also affects local–global processing. This exploration would contribute to a more comprehensive understanding of the relationship between analogy-solving and global–local processing.

Future research should be using other types of global–local processing tasks to examine whether solving distant analogies still could promote global processing. Both the Kimchi–Palmer task and the Navon letter task utilized in this study have been extensively validated through previous research as tools for assessing global–local processing. Notably, the Navon Letter task has been used in over 200 related studies (Isbell et al., 2016). However, Dale and Arnell (2013) found that the Kimchi–Palmer task has excellent measurement stability, while the Navon Letter task seems to have relatively poor measurement stability. More importantly, Dale and Arnell (2013) also found that each task may be measuring a unique aspect of global–local processing. Therefore, future studies should consider using other tasks that measure global–local processing (e.g., Global/local face task) for further examination.

Future studies should investigate whether individuals with a typical analytic cognitive style also process information globally when they solve distant analogies. Individuals with a holistic cognitive style have a preference for analyzing things as a whole and excel at identifying the global characteristics of objects. Conversely, individuals with an analytic cognitive style prefer to perceive things as collections of parts and excel at identifying the local features of items (Zhou et al., 2015). Therefore, individuals with a typical analytic cognitive style may not necessarily process information globally after solving distant analogies. However, further experimental evidence is necessary to corroborate this hypothesis.

Finally, solving distant analogies could promote global processing, a phenomenon observed in our current study, which holds some potential implications for non-laboratory contexts. In everyday life and the workplace, whether one is crafting a comprehensive travel plan or orchestrating a flawless corporate event, individuals must possess a keen sense of the broader perspective (holistic thinking). Therefore, encouraging individuals to regularly engage in solving distant analogies may prove advantageous in cultivating a holistic thinking approach. This, in turn, equips them with the capacity to excel in tasks reliant on holistic thinking. Besides, future research also could consider further investigating whether fostering holistic thinking can enhance individuals' spontaneous attention to abstract structural relationship similarities across various domains. Prior research, employing an in vivo approach in both scientific and industrial contexts, suggests that the exploration of structural relationships is more likely to result in creative and innovative outcomes during the idea-generation process (Dunbar & Blanchette, 2001; Green, 2016; Srinivasan et al., 2018). However, studies have demonstrated that, when it comes to problem-solving, both children and adults often exhibit limited spontaneous attention to structural relations (e.g., Simms & Richland, 2019). The present study found

that solving distant analogies could promote global processing. To elaborate, when individuals actively focus on abstract structural relational similarities for solving distant analogies, they tend to expand their perception and engage in more integrative thinking. Conversely, this finding suggests the intriguing possibility that people might personally benefit from embracing a “looking at the big picture” perspective, characterized by holistic thinking, as a response to their spontaneous attention to structural relations.

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Author contributions Dr. JL and Mr. KS conducted literature searches and provided summaries of previous research studies, designed the present study, and wrote the protocol. All the authors were involved in the fielding testing. Mr. TC conducted the statistical analysis, Mr. JG prepared Figs. 1, 2, 3, 4, and Miss. XW also conducted the statistical analysis. Dr. JL and Mr. KS wrote the draft of the manuscript and all authors contributed to and approved the final manuscript.

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Data availability We have uploaded the data to Open Science Framework. The specific website is https://osf.io/wb3yn/?view_only=0d54fe84fa3249e8b148b9279dd07ce5.

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest The authors declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Research involving human participants The study was conducted after obtaining Institutional Review Board approval from the Department of Psychology at Northwest Normal University. We received the written consent of all participants before testing began. All procedures performed in studies involving human participants were by the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

References

- Andrews, G., & Bohadana, G. R. (2018). Does solving distant analogies facilitate relational integration? *Journal of Cognitive Psychology*, 30(3), 270–280. <https://doi.org/10.1080/20445911.2017.1414223>
- Andrews, G., & Vann, D. M. (2019). Solving distant analogies reduces belief-based responding in transitive inference. *Journal of*

- Cognitive Psychology*, 31(7), 760–767. <https://doi.org/10.1080/20445911.2019.1657432>
- Barnett, S. M., & Ceci, S. J. (2002). When and where do we apply what we learn?: A taxonomy for far transfer. *Psychological Bulletin*, 128(4), 612–637. <https://doi.org/10.1037/0033-2909.128.4.612>
- Baumann, N., & Kuhl, J. (2005). Positive affect and flexibility: overcoming the precedence of global over local processing of visual information. *Motivation and Emotion*, 29(2), 123–134. <https://doi.org/10.1007/s11031-005-7957-1>
- Beilock, S. L., Rydell, R. J., & McConnell, A. R. (2007). Stereotype threat and working memory: Mechanisms, alleviation, and spillover. *Journal of Experimental Psychology: General*, 136(2), 256–276. <https://doi.org/10.1037/0096-3445.136.2.256>
- Burgoon, E. M., Henderson, M. D., & Markman, A. B. (2013). There are many ways to see the forest for the trees: A tour guide for abstraction. *Perspectives on Psychological Science*, 8(5), 501–520. <https://doi.org/10.1177/1745691613497964>
- Chaxel, A. (2015). The impact of a relational mindset on information distortion. *Journal of Experimental Social Psychology*, 60, 1–7. <https://doi.org/10.1016/j.jesp.2015.04.007>
- Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review*, 82(6), 407–428. <https://doi.org/10.1037/0033-295X.82.6.407>
- Dale, G., & Arnell, K. M. (2013). Investigating the stability of and relationships among global/local processing measures. *Attention, Perception, and Psychophysics*, 75, 394–406. <https://doi.org/10.3758/s13414-012-0416-7>
- Dale, G., & Arnell, K. M. (2015). Multiple measures of dispositional global/local bias predict attentional blink magnitude. *Psychological Research Psychologische Forschung*, 79, 534–547. <https://doi.org/10.1007/s00426-014-0591-3>
- De Luca, A., Verschoor, S., & Hommel, B. (2023). No correlation between mood or motivation and the processing of global and local information. *Experimental Psychology*. <https://doi.org/10.1027/1618-3169/a000562>
- Dijkstra, K. A., van der Pligt, J., van Kleef, G. A., & Kerstholt, J. H. (2012). Deliberation versus intuition: Global versus local processing in judgment and choice. *Journal of Experimental Social Psychology*, 48(5), 1156–1161. <https://doi.org/10.1016/j.jesp.2012.05.001>
- Dijkstra, K. A., Pligt, J., & Kleef, G. A. (2017). Fit between decision mode and processing style predicts subjective value of chosen alternatives. *European Journal of Social Psychology*, 47(1), 72–81. <https://doi.org/10.1002/ejsp.2217>
- Dunbar, K., & Blanchette, I. (2001). The in vivo/in vitro approach to cognition: The case of analogy. *Trends in Cognitive Sciences*, 5(8), 334–339. [https://doi.org/10.1016/S1364-6613\(00\)01698-3](https://doi.org/10.1016/S1364-6613(00)01698-3)
- Eid, M., Gollwitzer, M., & Schmitt, M. (2011). *Statistik und Forschungsmethoden Lehrbuch*. Weinheim: Beltz.
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G* Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>
- Franceschini, S., Bertoni, S., Puccio, G., Mancarella, M., Gori, S., & Facoetti, A. (2021). Local perception impairs the lexical reading route. *Psychological Research*, 85, 1748–1756. <https://doi.org/10.1007/s00426-020-01326-z>
- Gable, P. A., & Harmon-Jones, E. (2008). Approach-motivated positive affect reduces breadth of attention. *Psychological Science*, 19(5), 476–482. <https://doi.org/10.1111/j.1467-9280.2008.02112.x>
- Gable, P. A., Poole, B. D., & Harmon-Jones, E. (2015). Anger perceptually and conceptually narrows cognitive scope. *Journal of Personality and Social Psychology*, 109(1), 163–174. <https://doi.org/10.1037/a0039226>
- Gaspar, K., & Clore, G. L. (2002). Attending to the big picture: Mood and global versus local processing of visual information. *Psychological Science*, 13(1), 34–40. <https://doi.org/10.1111/1467-9280.00406>
- Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7(2), 155–170. [https://doi.org/10.1016/S0364-0213\(83\)80009-3](https://doi.org/10.1016/S0364-0213(83)80009-3)
- Gentner, D., & Maravilla, F. (2017). Analogical reasoning. In *International handbook of thinking and reasoning* (pp. 186–203). Routledge.
- Goldwater, M. B., & Jamrozik, A. (2019). Can a relational mind-set boost analogical retrieval? *Cognitive Research: Principles and Implications*, 4(1), 1–16. <https://doi.org/10.1186/s41235-019-0198-8>
- Gonen-Yaacovi, G., de Souza, L. C., Levy, R., Urbanski, M., Josse, G., & Volle, E. (2013). Rostral and caudal prefrontal contribution to creativity: a meta-analysis of functional imaging data. *Frontiers in Human Neuroscience*, 7, 465. <https://doi.org/10.3389/fnhum.2013.00465>
- Green, A. E. (2016). Creativity, within reason: Semantic distance and dynamic state creativity in relational thinking and reasoning. *Current Directions in Psychological Science*, 25(1), 28–35. <https://doi.org/10.1177/0963721415618485>
- Green, A. E., Kraemer, D. J., Fugelsang, J. A., Gray, J. R., & Dunbar, K. N. (2010). Connecting long distance: Semantic distance in analogical reasoning modulates frontopolar cortex activity. *Cerebral Cortex*, 20(1), 70–76. <https://doi.org/10.1093/cercor/bhp081>
- Green, A. E., Kraemer, D. J. M., Fugelsang, J. A., Gray, J. R., & Dunbar, K. N. (2012). Neural correlates of creativity in analogical reasoning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 38(2), 264–272. <https://doi.org/10.1037/a0025764>
- Gu, L., Yang, X., Li, L. M. W., Zhou, X., & Gao, D. (2017). Seeing the big picture: Broadening attention relieves sadness and depressed mood. *Scandinavian Journal of Psychology*, 58(4), 324–332. <https://doi.org/10.1111/sjop.12376>
- Guest, D., Gibbert, M., Estes, Z., Mazursky, D., & Lam, M. (2016). Modulation of taxonomic (versus thematic) similarity judgments and product choices by inducing local and global processing. *Journal of Cognitive Psychology*, 28(8), 1013–1025. <https://doi.org/10.1080/20445911.2016.1212057>
- Hansen, J., & Steinmetz, J. (2019). Motivated level of construal: How temperature affects the construal level of state-relevant stimuli. *Motivation and Emotion*, 43, 434–446. <https://doi.org/10.1007/s11031-018-09750-w>
- Henderson, M. D. (2013). When seeing the forest reduces the need for trees: The role of construal level in attraction to choice. *Journal of Experimental Social Psychology*, 49(4), 676–683. <https://doi.org/10.1016/j.jesp.2013.03.001>
- Holyoak, K. J., & Thagard, P. (1997). The analogical mind. *American Psychologist*, 52(1), 35–44. <https://doi.org/10.1037/0003-066X.52.1.35>
- Huntsinger, J. R., Clore, G. L., & Bar-Anan, Y. (2010). Mood and global-local focus: Priming a local focus reverses the link between mood and global-local processing. *Emotion*, 10(5), 722–726. <https://doi.org/10.1037/a0019356>
- Isbell, L. M., Rovenpor, D. R., & Lair, E. C. (2016). The impact of negative emotions on self-concept abstraction depends on accessible information processing styles. *Emotion*, 16(7), 1040–1049. <https://doi.org/10.1037/emo0000193>
- Ji, L., Yap, S., Best, M. W., & McGeorge, K. (2019). Global processing makes people happier than local processing. *Frontiers in Psychology*, 10, 670. <https://doi.org/10.3389/fpsyg.2019.00670>
- Kazakova, S., Cauberghe, V., Pandelaere, M., & De Pelsmacker, P. (2015). Can't see the forest for the trees? The effect of media multitasking on cognitive processing style. *Media Psychology*, 18(4), 425–450. <https://doi.org/10.1080/15213269.2015.1006789>
- Kimchi, R., & Palmer, S. E. (1982). Form and texture in hierarchically constructed patterns. *Journal of Experimental Psychology*:

- Human Perception and Performance*, 8(4), 521–535. <https://doi.org/10.1037/0096-1523.8.4.521>
- Kühnen, U., & Oyserman, D. (2002). Thinking about the self influences thinking in general: Cognitive consequences of salient self-concept. *Journal of Experimental Social Psychology*, 38(5), 492–499. [https://doi.org/10.1016/S0022-1031\(02\)00011-2](https://doi.org/10.1016/S0022-1031(02)00011-2)
- Li, J., Wang, F., Shen, M., & Fan, G. (2017). The tendency of unconscious thought toward global processing style. *Consciousness and Cognition: An International Journal*, 53, 14–22. <https://doi.org/10.1016/j.concog.2017.05.005>
- Li, J., Zhang, X., Zheng, H., Lu, Q., & Fan, G. (2019). Global processing styles facilitate the discovery of structural similarity. *Psychological Reports*, 122(5), 1755–1765. <https://doi.org/10.1177/003329411878749>
- Li, J., Shi, K., Wei, X., & Xia, Y. (2023a). Generating solutions to far analogies makes individuals tend to categorize information based on thematic relations. *Cognitive Processing*. <https://doi.org/10.1007/s10339-023-01148-y>
- Li, J., Shi, K., Guo, H., Sun, L., & Wang, S. (2023b). Global or local processing: relationship between multicultural experiences and information processing of minority group members. *Current Psychology*. <https://doi.org/10.1007/s12144-023-04541-0>
- Liberman, N., & Förster, J. (2009a). Distancing from experienced self: How global-versus-local perception affects estimation of psychological distance. *Journal of Personality and Social Psychology*, 97(2), 203–216. <https://doi.org/10.1037/a0015671>
- Liberman, N., & Förster, J. (2009b). The effect of psychological distance on perceptual level of construal. *Cognitive Science*, 33(7), 1330–1341. <https://doi.org/10.1111/j.1551-6709.2009.01061.x>
- Love, B. C., Rouders, J. N., & Wisniewski, E. J. (1999). A structural account of global and local processing. *Cognitive Psychology*, 38(2), 291–316. <https://doi.org/10.1006/cogp.1998.0697>
- Marguc, J., Förster, J., & Van Kleef, G. A. (2011). Stepping back to see the big picture: When obstacles elicit global processing. *Journal of Personality and Social Psychology*, 101(5), 883–901. <https://doi.org/10.1037/a0025013>
- Masuda, T., & Nisbett, R. E. (2001). Attending holistically versus analytically: Comparing the context sensitivity of Japanese and Americans. *Journal of Personality and Social Psychology*, 81(5), 922–934. <https://doi.org/10.1037/0022-3514.81.5.922>
- Mathis, K. M., & Kahan, T. A. (2014). Holistic processing improves change detection but impairs change identification. *Psychonomic Bulletin and Review*, 21, 1250–1254. <https://doi.org/10.3758/s13423-014-0614-5>
- Mok, A., & De Cremer, D. (2015). Overlooking interpersonal hurt: A global processing style influences forgiveness in work relationships. *European Journal of Work and Organizational Psychology*, 24(2), 267–278. <https://doi.org/10.1080/1359432X.2014.892872>
- Mukherjee, S., Srinivasan, N., Kumar, N., & Manjaly, J. A. (2018). Perceptual broadening leads to more prosociality. *Frontiers in Psychology*, 9, 10–19. <https://doi.org/10.3389/fpsyg.2018.01821>
- Navon, D. (1977). Forest before trees: The precedence of global features in visual perception. *Cognitive Psychology*, 9(3), 353–383. [https://doi.org/10.1016/0010-0285\(77\)90012-3](https://doi.org/10.1016/0010-0285(77)90012-3)
- Ng, H. K., Chen, S. X., Ng, J. C., & Ng, T. K. (2021). Does having multiple identities predict life satisfaction? Holistic thinking as a condition for achieving integrated self-concept. *Current Psychology*. <https://doi.org/10.1007/s12144-021-01477-1>
- Nisbett, R. E., & Miyamoto, Y. (2005). The influence of culture: Holistic versus analytic perception. *Trends in Cognitive Sciences*, 9(10), 467–473. <https://doi.org/10.1016/j.tics.2005.08.004>
- Poljac, E., de-Wit, L., & Wagemans, J. (2012). Perceptual wholes can reduce the conscious accessibility of their parts. *Cognition*, 123(2), 308–312. <https://doi.org/10.1016/j.cognition.2012.01.001>
- Schooler, J. W. (2002). Verbalization produces a transfer inappropriate processing shift. *Applied Cognitive Psychology*, 16(8), 989–997. <https://doi.org/10.1002/acp.930>
- Simms, N. K., & Richland, L. E. (2019). Generating relations elicits a relational mindset in children. *Cognitive Science*, 43(10), e12795. <https://doi.org/10.1111/cogs.12795>
- Smith, P. K., & Trope, Y. (2006). You focus on the forest when you're in charge of the trees: Power priming and abstract information processing. *Journal of Personality and Social Psychology*, 90(4), 578–596. <https://doi.org/10.1037/0022-3514.90.4.578>
- Srinivasan, N., & Hanif, A. (2010). Global-happy and local-sad: Perceptual processing affects emotion identification. *Cognition and Emotion*, 24(6), 1062–1069. <https://doi.org/10.1080/02699930903101103>
- Srinivasan, V., Song, B., Luo, J., Subburaj, K., Elara, M. R., Blessing, L., & Wood, K. (2018). Does analogical distance affect performance of ideation? *Journal of Mechanical Design*, 140(7), 071101. <https://doi.org/10.1115/1.4040165>
- Vendetti, M. S., Wu, A., & Holyoak, K. J. (2014). Far-out thinking: Generating solutions to distant analogies promotes relational thinking. *Psychological Science*, 25(4), 928–933. <https://doi.org/10.1177/0956797613518079>
- Viterbo, G., Katzir, T., & Goldfarb, L. (2020). Accelerating reading via local priming. *Acta Psychologica*, 205, 103056. <https://doi.org/10.1016/j.actpsy.2020.103056>
- Wan, M., Yang, Y., Liu, Y., & Li, J. (2016). Do multicultural experiences facilitate global processing style? *Asian Journal of Social Psychology*, 19(3), 209–214. <https://doi.org/10.1111/ajsp.12144>
- Ward, T. B. (1998). Analogical distance and purpose in creative thought: Mental leaps versus mental hops. In *Advances in analogy research: Integration of theory and data from the cognitive, computational, and neural sciences* (pp. 221–230)
- Wilford, M. M., & Wells, G. L. (2010). Does facial processing prioritize change detection? Change blindness illustrates costs and benefits of holistic processing. *Psychological Science*, 21(11), 1611–1615. <https://doi.org/10.1177/0956797610385952>
- Wronska, M. K., Kolańczyk, A., & Nijstad, B. A. (2018). Engaging in creativity broadens attentional scope. *Frontiers in Psychology*, 9, 1772. <https://doi.org/10.3389/fpsyg.2018.01772>
- Zhou, J., Zhou, C., Li, J., & Zhang, M. (2015). Cognitive style modulates conscious but not unconscious thought: Comparing the deliberation-without-attention effect in analytics and wholists. *Consciousness and Cognition*, 36(4), 54–60. <https://doi.org/10.1016/j.concog.2015.05.018>
- Zmigrod, S., Zmigrod, L., & Hommel, B. (2015). Zooming into creativity: Individual differences in attentional global-local biases are linked to creative thinking. *Frontiers in Psychology*, 6, 1647. <https://doi.org/10.3389/fpsyg.2015.01647>

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